Laboratory 3: Ballistic Trajectory with Drag

ENGN 0810

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Introduction

The purpose of this lab is to study the aerodynamics of a badminton shuttlecock by analyzing its motion under the effects of gravity and drag. When moving through a fluid, an object experiences a resistive force that acts opposite to its direction of motion: drag force. The drag force is dependent on the object's speed U, frontal area S, and drag coefficient C_d, which all vary with shape and fluid movement.

In this experiment, a badminton shuttlecock serves as a model for a high-drag projectile. Due to its light weight and conical shape, the shuttle experiences a large drag force that determines its flightpath. The motion of the shuttlecock is governed by the following equation:

$$M\frac{d\vec{U}}{dt} = M\vec{g} - \frac{1}{2}\rho U\vec{U}SC_D$$

The objective of this lab was to record and model the flight of a badminton shuttlecock at different launch speeds and angles, comparing experimental data against numerical predictions.

Methods

The experiment was conducted using a compressed-air cannon to launch badminton shuttlecocks at predetermined pressures and angles. The cannon, approximately 1.6 meters in length with a 5.1 centimeter barrel diameter, was connected to an air compressor pressurized to 50 psi.

A standard feathered shuttlecock with a drag coefficient of $C_d \sim 0.6$ was used for the baseline measurements. It was launched at three different angles: 32°, 45°, and 55°. Each trial was recorded using a smartphone camera positioned perpendicular to the flight path to capture the shuttle's trajectory. The cannon was included in the video to calibrate pixel measurements to physical distances.

The recorded video samples were then analyzed in MATLAB with image processing. The shuttle's position was extracted at multiple points in time to plot its horizontal and vertical trajectory. Additionally, simulations of the shuttlecock's motion were performed with a provided drag coefficient to generate predicted flight paths. Experimental results were then compared with these predictions.

For the design component of this experiment, a synthetic shuttlecock was constructed using various materials to attempt to mimic the general mass and shape of a feathered shuttle.

The main body of our shuttlecock was constructed with toothpicks inserted into a styrofoam base. The toothpicks were placed in a symmetrical conical pattern around the base. To replicate the 'goose feather' effect, a paper towel cut into a rhombus shape was hot glued onto each toothpick. This design is shown to the right.

The final piece of our assembly involved a rubber cap that was heat treated and trimmed to emulate the dome-like shape of a shuttlecock head. Our design weighed approximately 5.8 grams.

Figure 1: Top-down view of our shuttlecock design. Features a black rubber cap, toothpicks, and rhombus paper-feathers.

The synthetic shuttlecock was then launched under the same trial conditions as the real shuttlecock, allowing us to determine its trajectory and drag coefficient.

<u>Results</u>

Our MATLAB image analysis revealed the trajectories of the test shuttlecock and newly designed shuttlecock for each trial. Additionally, C_d values and exit velocities of the shuttles were extracted from the data. Two plots were created for each trial: one plot shows the y-position of the shuttlecock as a function of its x-position (*y vs. x)* and one plot shows the x-and-y positions as a function of time (*x or y vs. t)*. These plots are presented below for trials at 32°, 45°, and 55° with a constant pressure of 50 psi.

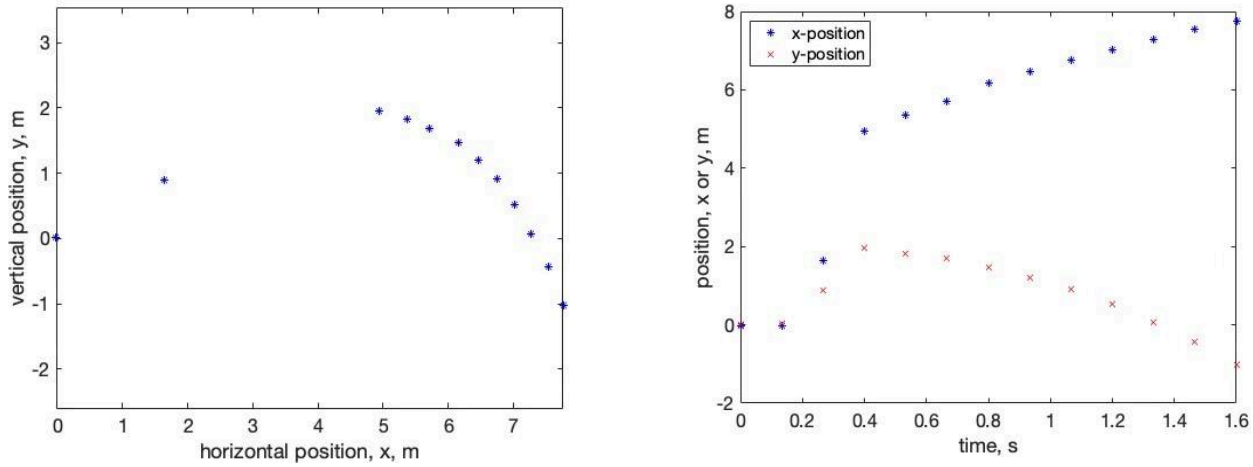

Figure 2: Plots of y vs. x (left) and x or y vs. t (right) for a test shuttlecock launched at 32° and 50 psi.

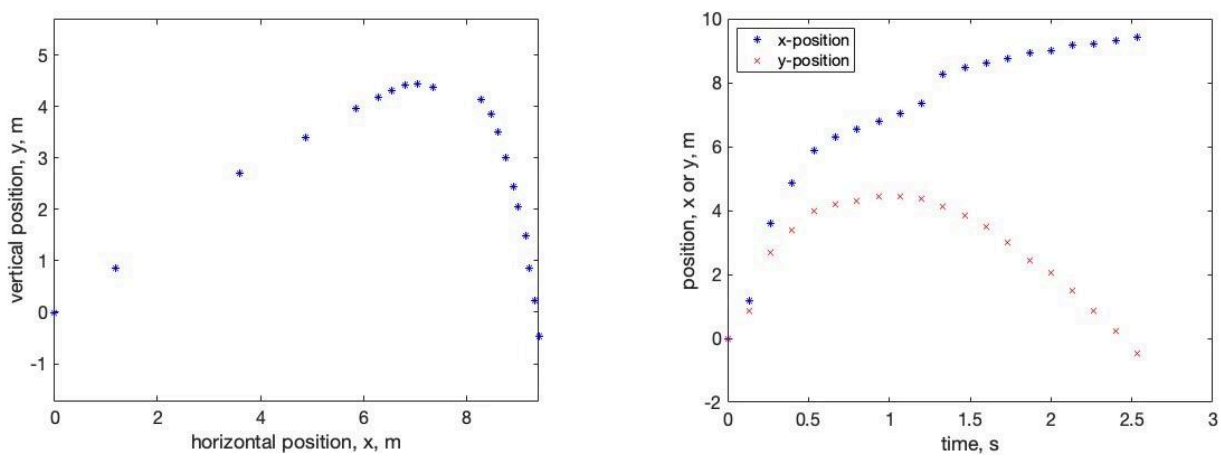

Figure 3: Plots of y vs. x (left) and x or y vs. t (right) for a newly designed shuttlecock launched at 32° and 50 psi.

In these 32° trials, the test shuttlecock exhibited an exit velocity v_o of 16.67 m/s and the new-design shuttlecock exhibited an exit velocity of 12.8 m/s. The new shuttlecock's C_d was found to be 0.2 for this trial. From this trial, it is evident that our experimental shuttlecock had a steeper trajectory than the test shuttlecock.

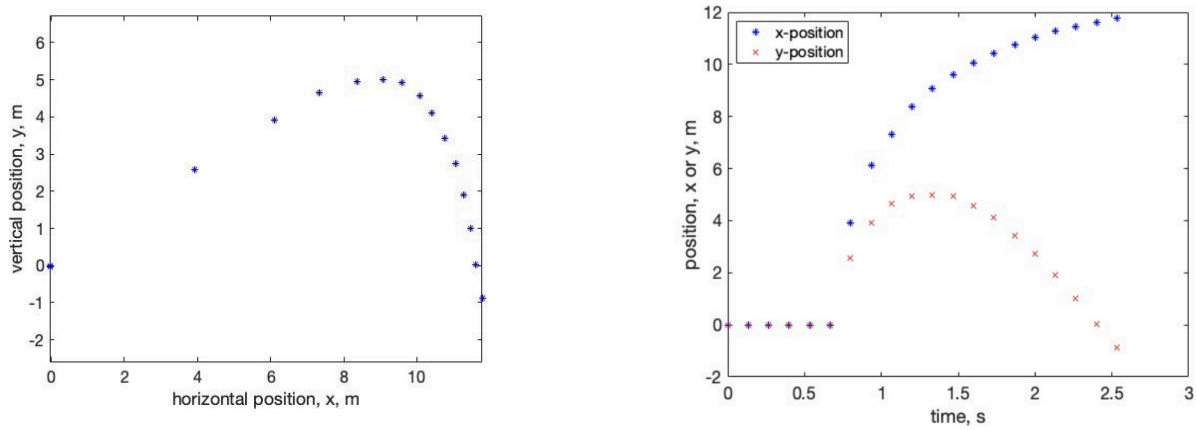

Figure 4: Plots of y vs. x (left) and x or y vs. t (right) for a test shuttlecock launched at 45° and 50 psi.

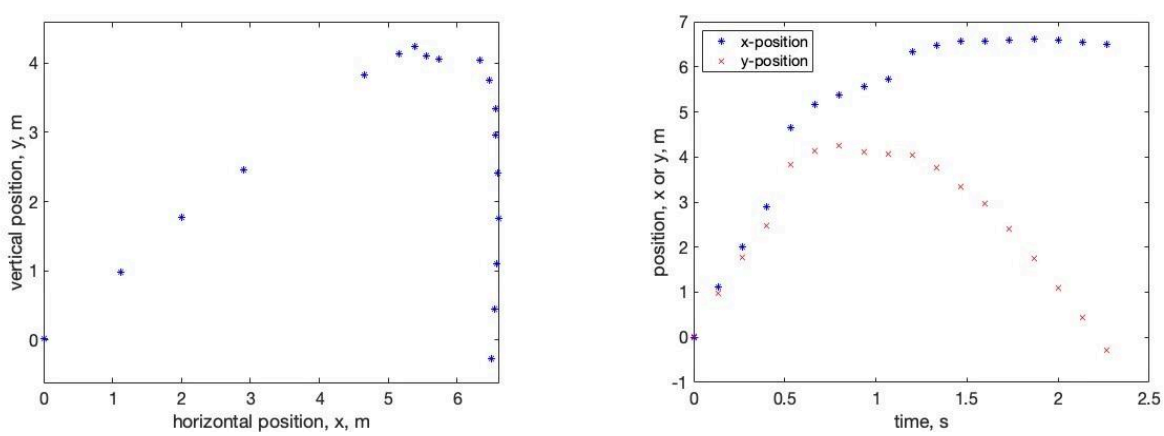

Figure 5: Plots of y vs. x (left) and x or y vs. t (right) for a newly designed shuttlecock launched at 45° and 50 psi.

In these 45° trials, the test shuttlecock once again exhibited an exit velocity (v_o) of 16.67 m/s and the new-design shuttlecock exhibited an exit velocity of 12.8 m/s. The new shuttlecock's C_d was found to be 0.4 for this trial. Similar to the previous trial at 32 degrees, the newly fabricated shuttlecock exhibited a steeper projectile curve than the test shuttlecock.

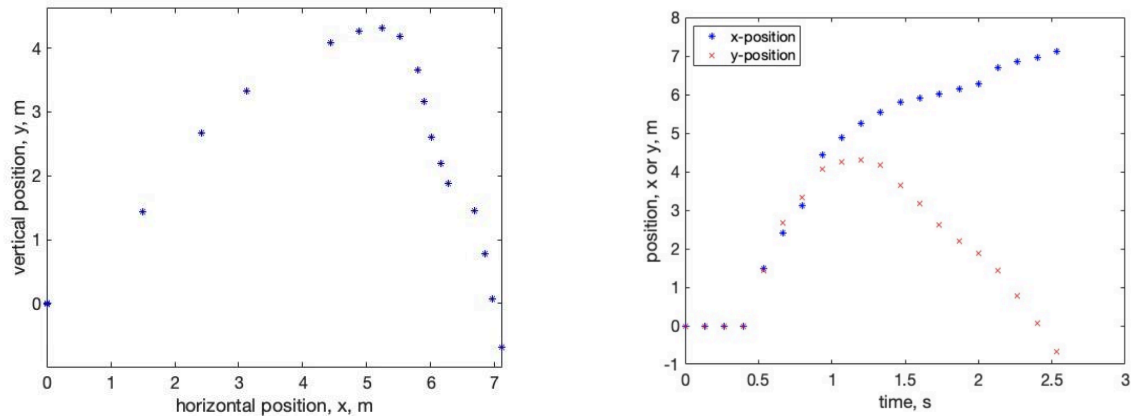

Figure 6: Plots of y vs. x (left) and x or y vs. t (right) for a newly designed shuttlecock launched at 55° and 50 psi.

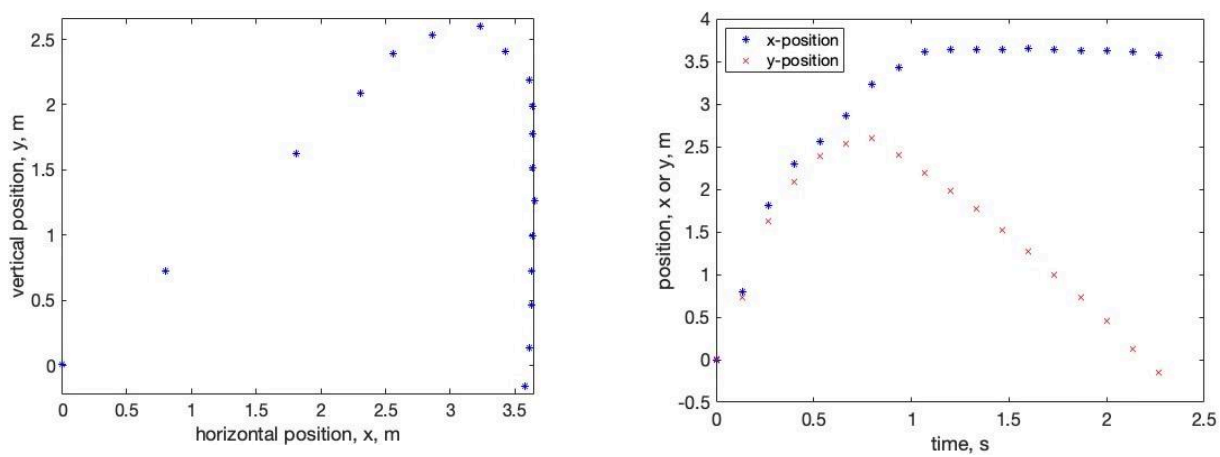

Figure 7: Plots of y vs. x (left) and x or y vs. t (right) for a newly designed shuttlecock launched at 55° and 50 psi.

In these 55° trials, the test shuttlecock once again exhibited an exit velocity v_o of 16.67 m/s and the new-design shuttlecock exhibited an exit velocity of 12.8 m/s. The new shuttlecock's C_d was found to be 0.9 for this trial. In this trial, it is apparent that once our shuttlecock hit a distance of ~3.6 m, it gained no further horizontal distance, unlike the test shuttlecock.

The C_d values for each of the trials were determined by fitting our experimental curves to a predicted curve of motion under the same conditions. Such an example is shown below:

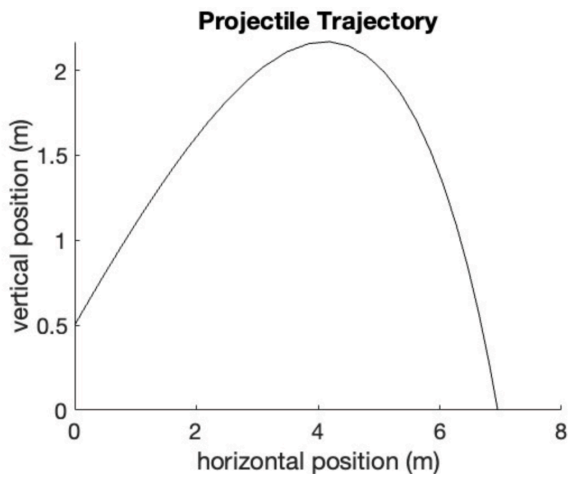

Figure 8: An experimental curve of a shuttlecock with a C_d of 0.6 launched at a pressure of 50 psi and an angle of 32°.

This graph provided a reference for our experimental shuttlecock trials.

Discussion, Conclusions, and Recommendations

Overall, the data collected from this experiment allowed us to draw several conclusions about our shuttlecock's design as well as the nature of projectile motion through a fluid. Primarily, we discovered three distinct C_d values in each of our experimental trials: at 32° our C_d was equal to 0.2, at 45° our C_d was equal to 0.4, and at 55° our C_d was equal to 0.9. Our average C_d over the three trials was equal to 0.5, which is near the known C_d value of 0.6 for the goose feather shuttlecock. However, given the large variability of C_d between trials, it is likely that our true C_d value is not precisely 0.5. In the future, a larger sample size of trials over the course of several days to control for external factors would provide greater accuracy in our measurements.

There were several factors that may have influenced the accuracy of our experimental results. While the setup successfully demonstrated the effects of drag on projectile motion, a number of procedural and environmental variables introduced varying amounts of uncertainty. Conducting the experiment outdoors limited control over extraneous variables such as wind conditions.

Given the relatively light weight of the shuttlecock, minor gusts and/or breezes had outsized effects on the flight path, and our data was collected on a gusty day. Additionally, the modified shuttlecock experienced rapid degradation after repeated launches, resulting in a decreased drag coefficient and increased flight distance. The materials and method of manufacturing in our design could not withstand the repeated force of the air cannon, unlike the goose feather shuttlecock.

Another limitation was the precision of the video analysis. In several frames across multiple videos, our shuttle was not visible, making its position difficult to precisely determine. In some instances, the shuttle also escaped the video frame, making its position and full trajectory impossible to determine precisely. Therefore, the accuracy of our experimental results was affected by the precision of the video analysis due to camera quality and unexpected trajectories of our designed shuttle.

Ultimately, while our design did not perfectly match the drag coefficient and trajectory of the goose feather shuttlecock, it produced a useful comparison to the known parameters of the manufacturer shuttlecock. We were able to discern three potential values for our drag coefficient and observed how our projectile's behavior changed at different angles and as the projectile deformed. Additionally, we observed how the goose feather shuttlecock was able to extract additional horizontal distance as the magnitude of its x-velocity decreased, which we were not able to replicate in our design. In future designs, we would consider this behavior to more closely match the performance of the manufactured shuttlecock.

To improve the lab for future students, we recommend a few changes. First, performing the launches indoors would be advantageous, as it would eliminate most of the error caused by outdoor externalities such as wind. Second, improving camera frame rate and video resolution would help to reduce uncertainty in video analysis. Perhaps the lab instructions could be changed to advise students to use the highest possible frame rate and video quality settings on their phone. Lastly, requiring students to perform extra trial launches with both the test and student-designed shuttles is advisable, as it would allow students to average out random errors better and provide more consistent results.

Appendix A: Error & Uncertainty Calculations of Drag Coefficient

Trial	Test C_d	Experimental C_d	Percent Error
θ = 32°, P = 50 psi	0.6	0.2	66.7%
θ = 32°, P = 50 psi	0.6	0.4	33.3%
θ = 32°, P = 50 psi	0.6	0.9	50.0%

Appendix B: Error & Uncertainty Calculations of Exit Velocity

Trial	Test Exit Velocity (v_o) in m/s	Experimental Exit Velocity (v_o) in m/s	Percent Error
θ = 32°, P = 50 psi	16.67	12.8	23.2%
θ = 45°, P = 50 psi	16.67	12.8	23.2%
θ = 55°, P = 50 psi	16.67	12.8	23.2%